1 Relative entropy for mixed excited states in Free Boson CFT

Setup and Replica Trick

We consider a massless free boson conformal field theory (CFT) in $1+1$ dimensions on a cylinder of circumference R. We focus on a spatial subsystem A defined by the interval $(-l/2, l/2)$ at a fixed time slice.

Our states of interest are the reduced density matrices corresponding to global excited coherent states. These states are generated by the action of vertex operators on the vacuum at past infinity:

$$|\alpha\rangle = V_\alpha|\Omega\rangle =: e^{i\alpha\phi} : |\Omega\rangle \tag{1.1}$$

The reduced density matrix of the state $|\alpha\rangle$ on the subsystem A is denoted by ρ_α.

We aim to compute the relative entropy $S(\rho||\sigma_y)$ where $\rho = \rho_\alpha$ and the reference state is the mixture $\sigma_y = y\rho_\alpha + (1-y)\rho_\beta$ for $0 \le y \le 1$. The standard replica trick for relative entropy relates this quantity to the analytic continuation of trace functionals that break replica symmetry. Specifically, for the mixed state case, we utilize the following identity derived from the generalized replica trick:

$$S\left(\rho_\alpha \middle\| y\rho_\alpha + (1-y)\rho_\beta\right) = \partial_n \log\left[\frac{\mathrm{tr}(\rho_\alpha^n)[\mathrm{tr}(y\rho_\alpha + (1-y)\rho_\beta)]^{n-1}}{\mathrm{tr}(\rho_\alpha(y\rho_\alpha + (1-y)\rho_\beta)^{n-1})[\mathrm{tr}(\rho_\alpha)]^{n-1}}\right]_{n\to 1} \tag{1.2}$$

The final answer will be (1.42).

The most non-trivial component of this calculation is the term $\mathrm{tr}(\rho(\rho+\sigma)^{n-1})$. Expanding this using the binomial theorem:This is incorrect because the operator order matters

$$\begin{aligned}\mathrm{tr}(\rho_\alpha(y\rho_\alpha + (1-y)\rho_\beta)^{n-1}) &= \sum_{p=0}^{n-1}\binom{n-1}{p} y^{n-1-p}(1-y)^p \mathrm{tr}(\rho_\alpha^{1+n-1-p}\rho_\beta^p) + \text{terms with commutators} \\ &= \sum_{p=0}^{n-1} y^{n-1-p}(1-y)^p \binom{n-1}{p}\mathrm{tr}(\rho_\alpha^{n-p}\rho_\beta^p) + \text{terms with commutators}\end{aligned} \tag{1.3}$$

Using the operator-state correspondence, the trace $\mathrm{tr}(\rho_\alpha^{n-p}\rho_\beta^p)$ corresponds to a path integral on an n-sheeted Riemann surface with specific operator insertions. In the limit $n \to 1$, we can identify this with the correlation function of p operators of type β and $q = n-p$ operators of type α:

$$C_{q,p} \equiv \langle \mathcal{O}_\alpha^{(q)} \mathcal{O}_\beta^{(p)} \rangle \tag{1.4}$$

As was pointed out, in the case of free bosons, the order o operators will not matter, and the commutator terms vanish.

Operator Insertions on the Strip

We perform the calculation on a strip of width 2π, obtained via the standard conformal mapping from the n-sheeted cylinder. The replica sheets are indexed by $k = 0, \ldots, n-1$.

The operator insertion points $t_k^{\pm}$ on the strip are:

$$t_k^{\pm} = \frac{2\pi k}{n} \pm \frac{\pi x}{n} \tag{1.5}$$

where $x = l/R$ is the dimensionless interval size. For the correlator $\langle \mathcal{O}_\alpha^{(q)} \mathcal{O}_\beta^{(p)} \rangle$ with $q = n - p$, the operators are explicitly distributed as follows: On sheets $k = 0, \ldots, q-1$ (α-sector) we insert V_α at t_k^+ and $V_{-\alpha}$ at t_k^-, and on sheets $k = q, \ldots, n-1$ (β-sector) we insert V_β at t_k^+ and $V_{-\beta}$ at t_k^-.

Calculation of the correlator

The correlation function for primary vertex operators $V_Q =: e^{iQ\phi} :$ is given by the product of distances raised to the power of the product of their charges. The chordal distance on the strip is $d(t_i, t_j) = 2\sin(\frac{t_i - t_j}{2})$.

$$\langle \prod_i V_{Q_i}(t_i) \rangle \sim \prod_{i<j} \left[2\sin\left(\frac{t_i - t_j}{2} \right) \right]^{Q_i Q_j} \tag{1.6}$$

- **Like charges** ($Q_i Q_j > 0$): The operators repel (regular zero), leading to a positive exponent $+Q_i Q_j$.
- **Opposite charges** ($Q_i Q_j < 0$): The operators attract (singular pole), leading to a negative exponent $-|Q_i Q_j|$.

We have the following contributions:

1. Nearest Neighbor Interactions (Same sheet, same sector) These are the contractions between t_k^+ and t_k^- on the same sheet.

- α-sector (q pairs): $(\alpha, -\alpha)$ separated by $\Delta t = 2\pi x/n$.

$$\Longrightarrow \left(2\sin\frac{\pi x}{n} \right)^{-q\alpha^2}$$

- β-sector (p pairs): $(\beta, -\beta)$ separated by $\Delta t = 2\pi x/n$.

$$\Longrightarrow \left(2\sin\frac{\pi x}{n} \right)^{-p\beta^2}$$

Putting the two together, we have

$$\mathcal{N}(q, p; x) = \left(2\sin\frac{\pi x}{n} \right)^{-(q\alpha^2 + p\beta^2)} \tag{1.7}$$

2. Intra-sector Interactions (Different sheets, same sector) For the α-sector, we sum over sheet separations $k = 1, \ldots, q-1$. The number of pairs at distance k is $(q-k)$.

- Like charges (α, α) and $(-\alpha, -\alpha)$:

$$\text{Dist} = \frac{2\pi k}{n} \implies \text{Exp} = +\alpha^2 \times 2$$

- Opposite charges $(\alpha, -\alpha)$: Two diagonals $\frac{2\pi k}{n} \pm \frac{2\pi x}{n}$.

$$\text{Dist} = \frac{2\pi(k \pm x)}{n} \implies \text{Exp} = -\alpha^2 \text{ (each)}$$

Collecting these terms for the α sector gives $[g(q;x)]^{\alpha^2}$:

$$g(q;x) = \prod_{k=1}^{q-1} \left(\frac{\sin^2(\frac{\pi k}{n})}{\sin(\frac{\pi(k+x)}{n}) \sin(\frac{\pi(k-x)}{n})} \right)^{q-k} . \tag{1.8}$$

The β sector is identical with $q \to p$ and $\alpha \to \beta$, and we have the overall contribution

$$g(q;x)^{\alpha^2} g(p;x)^{\beta^2} . \tag{1.9}$$

3. Cross-sector Interactions Interactions between the q α-sheets and p β-sheets yield terms involving the product $\alpha\beta$. These collect into $[h(p,q;x)]^{\alpha\beta}$:

$$h(q,p;x) = \prod_{l=1}^{p} \prod_{k=1}^{q} \frac{\sin^2(\frac{\pi(k+l-1)}{n})}{\sin(\frac{\pi(k+l-1+x)}{n}) \sin(\frac{\pi(k+l-1-x)}{n})} \tag{1.10}$$

Putting all the pieces together, we find that the full correlator in (1.4) is

$$\langle \mathcal{O}_\alpha^{(q)} \mathcal{O}_\beta^{(p)} \rangle = \mathcal{N}(n-p,p;x) \, [g(n-p;x)]^{\alpha^2} \, [g(p;x)]^{\beta^2} \, [h(n-p,p;x)]^{\alpha\beta} . \tag{1.11}$$

We need to continue in n analytically:

$$F(n) \equiv \text{tr}(\rho(\rho+\sigma)^{n-1}) = \sum_{p=0}^{n-1} \binom{n-1}{p} \langle \mathcal{O}_\alpha^{(q)} \mathcal{O}_\beta^{(p)} \rangle \tag{1.12}$$

1.1 Analytic continuation in n

We consider the relative entropy $S(\rho_\alpha || \sigma)$ where $\sigma = y\rho_\alpha + (1-y)\rho_\beta$ for excited coherent states in a 1+1D free boson CFT. The replica trick requires evaluating:

$$\text{tr}(\rho_\alpha \sigma^{n-1}) = \sum_{p=0}^{n-1} \binom{n-1}{p} y^{n-1-p} (1-y)^p C_{n-p,p} \tag{1.13}$$

where $C_{n-p,p} = \langle \mathcal{O}_\alpha^{(n-p)} \mathcal{O}_\beta^{(p)} \rangle$ is the mixed correlator on the n-sheeted replica surface.

Step 1 Log-Sum expansion of the correlator: ,k,1,The correlator is a product of geometric functions g and h. We focus on the intra-sector term $\log g(p;x)$. For integer p, it is defined as:

$$\log g(p;x) = \sum_{k=1}^{p-1}(p-k)\left[2\log d\left(\frac{k}{n}\right) - \log d\left(\frac{k+x}{n}\right) - \log d\left(\frac{k-x}{n}\right)\right] \tag{1.14}$$

where $d(y) = 2\sin(\pi y)$ is the distance between operator insertions. We utilize the Fourier expansion for $\log\sin(\pi y)$:

$$\log\sin(\pi y) = -\log 2 - \sum_{m=1}^{\infty}\frac{\cos(2\pi m y)}{m} \tag{1.15}$$

Substituting this into the potential $\mathcal{V}(k,x)$ inside the sum, the constants cancel, yielding:

$$\mathcal{V}(k,x) = -2\sum_{m=1}^{\infty}\frac{1}{m}\left[1-\cos\left(\frac{2\pi m x}{n}\right)\right]\cos\left(\frac{2\pi m k}{n}\right) \tag{1.16}$$

The expression above has the advantage that it has separated the contribution of k from those of x, and we can now perform the sum over k.

Step 2 Sum over k: Following Section 6 of Calabrese, Cardy, and Tonni (2011), we perform the sum over k for each mode m:

$$\log g(p;x) = -2\sum_{m=1}^{\infty}\frac{1-\cos(2\pi m x/n)}{m}\sum_{k=1}^{p-1}(p-k)\cos\left(\frac{2\pi m k}{n}\right) \tag{1.17}$$

The inner sum $S_m(p) = \sum_{k=1}^{p-1}(p-k)\cos(k\theta_m)$ is evaluated using geometric series. For integer p and $\theta_m = 2\pi m/n$:

$$S_m(p) = \mathrm{Re}\left[\frac{pe^{i\theta_m}}{1-e^{i\theta_m}} - \frac{e^{i\theta_m}(1-e^{ip\theta_m})}{(1-e^{i\theta_m})^2}\right] = \frac{-p}{2} + \frac{(1-\cos(p\theta_m))}{4\sin^2(\theta_m/2)}, \tag{1.18}$$

Therefore, we find

$$\log g(p;x) = \sum_{m=1}^{\infty}\frac{1-\cos(\frac{2\pi m x}{n})}{m}\left(p - \frac{\sin^2(\pi p m/n)}{\sin^2(\pi m/n)}\right) \tag{1.19}$$

Define the mode kernels

$$L(n,x) := \sum_{m=1}^{\infty}\frac{1-\cos(2\pi m x/n)}{m}, \qquad K_m(n,x) := \frac{1-\cos(2\pi m x/n)}{2m\,\sin^2(\pi m/n)}. \tag{1.20}$$

to write

$$\log g(p;x) = p\,L(n,x) \;-\; \sum_{m=1}^{\infty}K_m(n,x)\,\big(1-\cos(p\theta_m)\big). \tag{1.21}$$

All p-independent pieces can be absorbed into $\Omega_n^{(\mathrm{rest})}(x)$ later.

A technical note to keep in mind is that to make every rearrangement fully justified, one may define all mode sums with a common Abel factor $e^{-\varepsilon m}$ in the summand and take $\varepsilon \to 0^+$ at the end. This guarantees absolute and uniform convergence in intermediate steps.

Next, we move to the function $h(q,p;x)$ in (1.10) which includes a double product over (k,l) with $1 \le k \le q$, $1 \le l \le p$. Taking logs and using the same log-sum representation as in we did for the g function yields

$$\log h(q,p;x) = \sum_{r=1}^{n-1} w_r(p,q)\, V(r,x), \tag{1.22}$$

where $w_r(p,q)$ counts the number of pairs (k,l) such that $k+l-1=r$.

It is convenient to evaluate the weighted cosine sum via generating functions. Let θ be arbitrary and note

$$\sum_{r=1}^{n-1} w_r(p,q) e^{\mathrm{i}r\theta} = \sum_{k=1}^{q}\sum_{l=1}^{p} e^{\mathrm{i}(k+l-1)\theta} = e^{-\mathrm{i}\theta}\Big(\sum_{k=1}^{q} e^{\mathrm{i}k\theta}\Big)\Big(\sum_{l=1}^{p} e^{\mathrm{i}l\theta}\Big). \tag{1.23}$$

Now specialize to the case $q = n-p$ and $\theta = \theta_m = 2\pi m/n$. Using $\sum_{k=1}^{n-p} e^{\mathrm{i}k\theta_m} = -e^{-\mathrm{i}p\theta_m/2}\,\frac{\sin(p\theta_m/2)}{\sin(\theta_m/2)}$ and $\sum_{l=1}^{p} e^{\mathrm{i}l\theta_m} = e^{\mathrm{i}(p+1)\theta_m/2}\,\frac{\sin(p\theta_m/2)}{\sin(\theta_m/2)}$, the right-hand side of (1.23) becomes real and equals

$$\sum_{r=1}^{n-1} w_r(p,n-p)\cos(r\theta_m) = -\left(\frac{\sin(p\theta_m/2)}{\sin(\theta_m/2)}\right)^2 = -\frac{1-\cos(p\theta_m)}{2\sin^2(\theta_m/2)}. \tag{1.24}$$

Substituting (1.24) into (1.22) gives the exact mode form

$$\log h(n-p,p;x) = 2\sum_{m=1}^{\infty} K_m(n,x)\,\big(1-\cos(p\theta_m)\big), \tag{1.25}$$

with the *same* $K_m(n,x)$ as in (1.20).

We are now ready to put together all the pieces back into $C_{n-p,p}$. Using (1.21) and (1.25), we find

$$\begin{aligned}\alpha^2\log g(q;x) + \beta^2\log g(p;x) + \alpha\beta\log h(q,p;x) &= (\alpha^2 q + \beta^2 p)\,L(n,x)\\ &\quad - (\alpha-\beta)^2\sum_{m=1}^{\infty} K_m(n,x)\,\big(1-\cos(p\theta_m)\big)\end{aligned} \tag{1.26}$$

where we have used a similar log-sum expansion for the function $\mathcal{N}$. Absorbing all p-independent constants (including the $-(\alpha-\beta)^2\sum_m K_m$ term) into $\Omega_n(x)$, we obtain

$$C_{n-p,p}(x) = \Omega_n(x)\,\Lambda_\alpha^{n-p}\,\Lambda_\beta^{p}\,\exp\left[\sum_{m=1}^{\infty} a_m(n,x)\cos(p\theta_m)\right], \tag{1.27}$$

where

$$a_m(n,x) = (\alpha-\beta)^2\,K_m(n,x) = (\alpha-\beta)^2\,\frac{1-\cos(2\pi m x/n)}{2m\,\sin^2(\pi m/n)}\ . \tag{1.28}$$

1.2 Fourier/Bessel expansion and definition of the R-modes

Let $\theta := \frac{2\pi p}{n}$ so that $\cos(p\theta_m) = \cos(m\theta)$. Using $e^{a\cos(m\theta)} = \sum_{r\in\mathbb{Z}} I_r(a)\, e^{\mathrm{i}rm\theta}$ (modified Bessel functions), we expand

$$\exp\left[\sum_{m\geq 1} a_m \cos(m\theta)\right] = \sum_{R\in\mathbb{Z}} W_R(n,x;\alpha,\beta)\, e^{\mathrm{i}R\theta}, \tag{1.29}$$

where the coefficients are the (infinite) Bessel convolution

$$W_R(n,x;\alpha,\beta) = \sum_{\{r_m\}:\sum_{m\geq 1} m r_m = R} \prod_{m\geq 1} I_{r_m}\big(a_m(n,x)\big), \qquad W_{-R} = W_R. \tag{1.30}$$

This allows us to rewrite

$$C_{n-p,p}(x) = \Omega_n\, \Lambda_\alpha^{n-p}\Lambda_\beta^{p} \sum_{R\in\mathbb{Z}} W_R(n)\, e^{\mathrm{i}\frac{2\pi R}{n}p}. \tag{1.31}$$

Note that with an Abel regulator in the defining mode sums, the function of θ in (1.29) is real-analytic in a complex strip, which implies exponential decay of W_R and in particular $\sum_R |W_R| < \infty$.

The expression in (1.31) has the advantage that if we can swap the sum over R with the sum over p, we can perform the p-sum explicitly. That is our next step. **I believe swapping the two sums can be justified using Abel regulators but I am not entirely sure!**

Plugging these back in 1.12 and swapping the order of summation, we obtain

$$\begin{aligned} F(n) &= \Omega_n\, \Lambda_\alpha \sum_{R\in\mathbb{Z}} W_R(n) \sum_{p=0}^{n-1} \binom{n-1}{p} (y\Lambda_\alpha)^{n-1-p} \left((1-y)\Lambda_\beta e^{\mathrm{i}2\pi R/n}\right)^p \\ &= \Omega_n\, \Lambda_\alpha \sum_{R\in\mathbb{Z}} W_R(n) \left(y\Lambda_\alpha + (1-y)\Lambda_\beta e^{\mathrm{i}2\pi R/n}\right)^{n-1}. \end{aligned} \tag{1.32}$$

Next, we would like to argue that only the $R=0$ term in the expression above contrinutes to the $n\to 1$ limit. To establish this, define $\varepsilon = n-1$ and

$$A_R(n) := y\Lambda_\alpha + (1-y)\Lambda_\beta e^{\mathrm{i}2\pi R/n}, \qquad A_0 := y\Lambda_\alpha + (1-y)\Lambda_\beta > 0. \tag{1.33}$$

Then (1.32) reads

$$F(n) = \Omega_n \Lambda_\alpha \sum_{R\in\mathbb{Z}} W_R(n)\, A_R(n)^{\varepsilon}. \tag{1.34}$$

With an Abel regulator W_R decays exponentially and has finite moments. Expand the phase for $n = 1+\varepsilon$:

$$e^{\mathrm{i}2\pi R/n} = e^{\mathrm{i}2\pi R/(1+\varepsilon)} = \exp\big(-\mathrm{i}2\pi R\varepsilon + O(R\varepsilon^2)\big), \tag{1.35}$$

so $A_R(n) = A_0 + \delta_R$ with $\delta_R = O(R\varepsilon)$ for fixed R and $\delta_{-R} = \overline{\delta_R}$. Using

$$A_R^\varepsilon = \exp\bigl(\varepsilon \log(A_0 + \delta_R)\bigr) = A_0^\varepsilon \left[1 + \varepsilon \frac{\delta_R}{A_0} + O(\varepsilon |\delta_R|^2)\right], \tag{1.36}$$

and the evenness $W_{-R} = W_R$, the potentially dangerous linear term $\sum_R W_R\, \delta_R$ cancels at $O(\varepsilon)$ (its leading part is odd/purely imaginary), leaving only $O(\varepsilon^2)$ corrections after summing over R. Consequently, after absorbing $\sum_R W_R$ into Ω_n (a p-independent normalization),

$$F(1+\varepsilon) = \Omega_{1+\varepsilon} \Lambda_\alpha\, A_0^\varepsilon \bigl(1 + O(\varepsilon^2)\bigr). \tag{1.37}$$

Thus the $R \neq 0$ modes do not affect $\partial_n \log F(n)$ at $n = 1$.

The replica prescription for the relative entropy is

$$S(\rho_\alpha \| \sigma_y) = \partial_n \Bigl(\log \operatorname{tr}(\rho_\alpha^n) - \log \operatorname{tr}(\rho_\alpha \sigma_y^{n-1})\Bigr)\Big|_{n=1}. \tag{1.38}$$

In the same framework one has $\operatorname{tr}(\rho_\alpha^n) = \widetilde{\Omega}_n\, \Lambda_\alpha^n$ so that $\partial_n \log \operatorname{tr}(\rho_\alpha^n)|_{n=1} = \log \Lambda_\alpha$ (all $\widetilde{\Omega}_n$ terms cancel against Ω_n in the difference). Using (1.37),

$$\partial_n \log \operatorname{tr}(\rho_\alpha \sigma_y^{n-1})\Big|_{n=1} = \log A_0 = \log\bigl(y\Lambda_\alpha + (1-y)\Lambda_\beta\bigr). \tag{1.39}$$

Therefore

$$S(\rho_\alpha \| \sigma_y) = \log \Lambda_\alpha - \log\bigl(y\Lambda_\alpha + (1-y)\Lambda_\beta\bigr) = -\log\left[y + (1-y)\frac{\Lambda_\beta}{\Lambda_\alpha}\right]. \tag{1.40}$$

For coherent-state excitations in the free boson CFT one may identify

$$\frac{\Lambda_\beta}{\Lambda_\alpha} = e^{-S(\alpha\|\beta)}, \qquad S(\alpha\|\beta) = (\alpha-\beta)^2 K(x), \qquad K(x) = 1 - \pi x \cot(\pi x). \tag{1.41}$$

Substituting (1.41) into (1.40) gives the final answer

$$\boxed{S\Bigl(\rho_\alpha \,\Big\|\, y\rho_\alpha + (1-y)\rho_\beta\Bigr) = -\log\bigl(y + (1-y)\exp\bigl[-(\alpha-\beta)^2(1-\pi x\cot(\pi x))\bigr]\bigr)\,.} \tag{1.42}$$

2 Consistency tests

Let $s := (\alpha-\beta)^2 K(x) \geq 0$ and $a := e^{-s} \in (0,1]$ so that

$$S(y) \equiv S\Bigl(\rho_\alpha \,\Big\|\, y\rho_\alpha + (1-y)\rho_\beta\Bigr) = -\log\,(y + (1-y)a) = -\log(a + (1-y)a) \tag{2.1}$$

We are gonna test the following limits:

- **Identical states:** $\alpha = \beta \Rightarrow s = 0 \Rightarrow a = 1 \Rightarrow S(y) = 0$ as expected.
- **No mixing:** $S(1) \to 0$ as expected and $S(0) = S(\alpha\|\beta)$ as expected. Therefore, we can write

$$e^{-S(y)} = y e^{-S(1)} + (1-y) e^{-S(0)}\,. \tag{2.2}$$

- **Much heavier state:** $s \to \infty \Rightarrow a \to 0 \Rightarrow S_y \to -\log y$.

Next, we test the following properties:

- **Positivity:** Since $y \leq a + y(1-a) \leq 1$, we have $0 \leq S(y) \leq -\log y$ as required fo r relative entropy.

- **Convexity in y:** Taking derivatives with respect to y

$$S'(y) = -\frac{1-a}{a+y(1-a)} \leq 0, \qquad S''(y) = \frac{(1-a)^2}{(a+y(1-a))^2} \geq 0. \tag{2.3}$$

  Thus $S(y)$ is decreasing and convex in the mixing parameter y (strictly convex if $s > 0$), consistent with the general theorem that $S(\rho\|\sigma)$ is convex in σ.

- **Monotonicity and concavity in s:** Treating S as a function of s:

$$\frac{\partial S}{\partial s} = \frac{(1-y)e^{-s}}{y+(1-y)e^{-s}} \geq 0, \qquad \frac{\partial^2 S}{\partial s^2} = -\frac{y(1-y)e^{-s}}{(y+(1-y)e^{-s})^2} \leq 0. \tag{2.4}$$

  So S increases with s but is concave in s, consistent with saturation at $-\log y$. A direct derivative gives the derivative of relative entropy in region size

$$K'(x) = \frac{\pi}{\sin^2(\pi x)}\left(\pi x - \frac{1}{2}\sin(2\pi x)\right) > 0 \qquad \text{for } x \in (0,1), \tag{2.5}$$

  since $\sin u < u$ for $u > 0$. Thus $K(x)$ (and therefore S) is increasing in x for $\alpha \neq \beta$, as expected from data processing.

Finally, we note that in the small interval limit $x \ll 1$ we have

$$K(x) = 1 - \pi x \cot(\pi x) = \frac{\pi^2}{3}x^2 + \frac{\pi^4}{45}x^4 + O(x^6). \tag{2.6}$$

Hence

$$S(y) = (1-y)\,(\alpha-\beta)^2\frac{\pi^2}{3}x^2 + O(x^4). \tag{2.7}$$

The relative entropy vanishes as x^2 for small intervals, as expected.

Alternative analytic continuation

An alternative analytic continuation follows from observing that since we are interested in small $n-1$, and the sum over $p = 1, \cdots, n-1$, we linearize our functions in p and discard higher order terms:

$$S_m(p) = -\frac{p}{2} + \frac{1-\cos(p\theta_m)}{4\sin^2(\theta_m/2)} = -\frac{p}{2} + O(p^2). \tag{2.8}$$

Expanding the g function in $n = 1+\epsilon$ results in

$$\log g(p;x) = p\,L(n,x) + O(p^2) = p\,L(n,x) + O(\varepsilon^2), \tag{2.9}$$

uniformly over the relevant range $p \in \{0, \ldots, n-1\}$. This reproduces the linear-in-p structure that one would obtain by keeping only the "smooth" term $S_m(p) \approx -p/2$ and effectively discarding the oscillatory $\cos(p\theta_m)$ piece at this order. While this results in a different analytic continuation than what we computed above, the $n \to 1$ limit gives the same result, and the same final expression (1.42) for relative entropy.

Pole-matching analytic continuation of $g(p;x)$ and $h(n-p,p;x)$

Here, we reformulate the analytic continuation of the building blocks $g(p;x)$ and $h(n-p,p;x)$ using the method used in the appendix of `arXiv:1508.03506`. The idea is to this: To find the analytic cotinuation of a meromorphic function $f_1(z)$, you find another entire function with the same poles and zeros $f_2(z)$ such that $f_1(z)/f_z(2)$ is bounded. Then, by Liouville's theorem it must be a constant C. This means that $Cf_2(z)$ is the analytic continuation of $f_1(z)$.

The basic building block and the definitions

Introduce

$$f_{n,m}(x) := \frac{\sin^2(\pi m/n)}{\sin\big(\pi(m+x)/n\big)\,\sin\big(\pi(m-x)/n\big)}\,, \qquad m \in \mathbb{Z}. \tag{2.10}$$

For integer $n \geq 2$ and $p \in \{1, \ldots, n-1\}$, the functions appearing in the correlator are

$$g(p;x) = \prod_{m=1}^{p-1} f_{n,m}(x)^{\,p-m}, \tag{2.11}$$

$$h(q,p;x) = \prod_{l=1}^{p}\prod_{k=1}^{q} f_{n,k+l-1}(x) = \prod_{r=1}^{n-1} f_{n,r}(x)^{\,w_r(p,q)}, \qquad q = n-p, \tag{2.12}$$

where $w_r(p,q)$ counts the number of pairs (k,l) such that $k+l-1=r$.

Poles of $g(p;x)$ as a function of x

From (2.11), all x-dependence sits in the denominator sines. For fixed m, the factor $\sin(\pi(m \pm x)/n)$ vanishes iff $m \pm x = \ell n$ for some $\ell \in \mathbb{Z}$, i.e.

$$x = \pm m + \ell n. \tag{2.13}$$

Hence the pole set of $g(p;x)$ is

$$\boxed{\mathrm{Pole}(g) = \Big\{\, x = \pm m + \ell n \;\Big|\; m = 1, \ldots, p-1,\ \ell \in \mathbb{Z} \Big\}.} \tag{2.14}$$

Moreover, each vanishing sine gives a simple pole, and the factor $f_{n,m}(x)$ is raised to power $(p-m)$. Therefore the *pole order* of $g(p;x)$ at $x = \pm m + \ell n$ equals $(p-m)$:

$$\boxed{\mathrm{ord}_{x=\pm m+\ell n}\, g(p;x) = p - m.} \tag{2.15}$$

Finally, $g(p;x)$ has no zeros as a function of x, since the numerator $\sin^2(\pi m/n)$ is x-independent.

A multiple-sine candidate and its divisor

We now introduce a candidate $g_{\mathrm{MS}}(p;x)$ built from multiple sine functions. Let $S_2(z|1,n)$ and $S_3(z|1,1,n)$ denote the double and triple sine functions (in a standard normalization), characterized by the shift relations

$$\frac{S_2(z+1\,|\,1,n)}{S_2(z\,|\,1,n)} = 2\sin\!\Big(\frac{\pi z}{n}\Big), \qquad \frac{S_3(z+1\,|\,1,1,n)}{S_3(z\,|\,1,1,n)} = \frac{1}{S_2(z\,|\,1,n)}. \tag{2.16}$$

Define the triangular-product package

$$T_p(z) := \frac{S_3(z+2\,|\,1,1,n)^{\,p}}{S_3(z+p+1\,|\,1,1,n)\,S_3(z+1\,|\,1,1,n)^{\,p-1}}, \tag{2.17}$$

and the candidate analytic continuation

$$g_{\mathrm{MS}}(p;x) := \frac{T_p(0)^2}{T_p(x)\,T_p(-x)}. \tag{2.18}$$

Claim (telescoping to the finite sine product). By iterating the shift relations (2.16), one checks that $T_p(z)$ is proportional to the triangular sine product

$$T_p(z) \;\propto\; \prod_{m=1}^{p-1} \sin\!\Big(\frac{\pi(m+z)}{n}\Big)^{p-m}, \tag{2.19}$$

where the proportionality factor is independent of z (it may depend on n,p). Consequently, (2.18) implies that for integer n, $g_{\mathrm{MS}}(p;x)$ reproduces the same pole lattice and the same pole orders as $g(p;x)$.

Zeros/poles of $g_{\mathrm{MS}}(p;x)$. From (2.19), $T_p(z)$ has zeros whenever $z = \ell n - m$ for $m = 1,\dots,p-1$, with zero order $(p-m)$. Therefore $g_{\mathrm{MS}}(p;x)$ has poles whenever $T_p(x)=0$ or $T_p(-x)=0$, i.e.

$$x = \ell n - m \quad \text{or} \quad x = -(\ell n - m) = m - \ell n \quad \Longleftrightarrow \quad x = \pm m + \ell n, \tag{2.20}$$

and the pole order at each such point equals $(p-m)$. In particular,

$$\boxed{\mathrm{Pole}(g_{\mathrm{MS}}) = \mathrm{Pole}(g), \qquad \mathrm{ord}_{x=\pm m+\ell n}\, g_{\mathrm{MS}}(p;x) = p-m.} \tag{2.21}$$

Like g, the function g_{MS} has no zeros as a function of x.

The ratio has no poles and is bounded

Define the ratio

$$R(x) := \frac{g_{\mathrm{MS}}(p;x)}{g(p;x)}. \tag{2.22}$$

Since g_{MS} and g have the same poles with the same orders and no zeros, all singularities cancel:

$$\boxed{R(x) \text{ is entire in } x.} \tag{2.23}$$

Moreover, both g and g_{MS} are even and n-periodic in x, hence so is R:

$$R(-x) = R(x), \qquad R(x+n) = R(x). \tag{2.24}$$

To argue boundedness, it suffices (by periodicity) to bound R on a single vertical strip $\{x = u + \mathrm{i}v : u \in [0, n]\}$. On such a strip, the original $g(p; x)$ obeys an exponential-type estimate as $|v| \to \infty$: for each fixed m,

$$\Big|\sin\Big(\frac{\pi(m \pm (u + \mathrm{i}v))}{n}\Big)\Big| \sim \frac{1}{2} e^{\pi|v|/n}, \qquad |v| \to \infty, \tag{2.25}$$

so

$$|g(p; u + \mathrm{i}v)| \sim C \exp\Big[-\frac{2\pi|v|}{n} \sum_{m=1}^{p-1} (p - m)\Big] = C \exp\Big[-\frac{\pi p(p-1)}{n}|v|\Big], \qquad |v| \to \infty, \tag{2.26}$$

with C independent of v (uniformly for u in compact sets away from poles).

On the other hand, the multiple sine functions S_2, S_3 have well-known large-imaginary asymptotics of the form

$$\log S_r(u + \mathrm{i}v \,|\, \boldsymbol{\omega}) = \pm \frac{\pi}{r!\,\omega_1 \cdots \omega_r} |v|^r \;+\; \text{lower powers in } |v| \;+\; O(1), \qquad |v| \to \infty, \tag{2.27}$$

so that in ratios such as (2.17) the leading polynomial growth cancels, and $T_p(z)$ has exponential type matching the finite sine product in (2.19). Consequently $g_{\mathrm{MS}}(p; u + \mathrm{i}v)$ has the same exponential type as $g(p; u + \mathrm{i}v)$, and hence their ratio $R(u + \mathrm{i}v)$ remains bounded as $|v| \to \infty$ uniformly in $u \in [0, n]$. Together with continuity on the compact region $\{u \in [0, n],\ |v| \le V\}$, we conclude:

$$\boxed{R(x) \text{ is bounded on } \mathbb{C}.} \tag{2.28}$$

Since R is entire and bounded, Liouville's theorem implies R is constant. Finally, $g(p; 0) = 1$ by definition and $g_{\mathrm{MS}}(p; 0) = 1$ by (2.18), so $R(0) = 1$ and thus

$$\boxed{g_{\mathrm{MS}}(p; x) = g(p; x) \quad \text{for all } x \text{ (at integer } n).} \tag{2.29}$$

This is precisely the "same poles + bounded ratio" strategy.

Corollary: $h(n-p, p; x)$ from a full-product identity

Using (2.12) with $q = n - p$ and the explicit multiplicities

$$w_r(p, n-p) = \begin{cases} r, & 1 \le r \le p, \\ p, & p < r \le n - p, \\ n - r, & n - p < r \le n - 1, \end{cases} \tag{2.30}$$

one may rewrite $h(n-p, p; x)$ as

$$h(n-p, p; x) = \Big(\prod_{r=1}^{n-1} f_{n,r}(x)\Big)^p g(p; x)^{-2}. \tag{2.31}$$

The appendix identity of `arXiv:1508.03506` proves

$$\prod_{r=1}^{n-1} f_{n,r}(x) = \left(\frac{n\sin(\pi x/n)}{\sin(\pi x)}\right)^2, \tag{2.32}$$

again by the same poles/bounded ratio argument. Substituting into (2.31) yields

$$\boxed{h(n-p,p;x) = \left(\frac{n\sin(\pi x/n)}{\sin(\pi x)}\right)^{2p} g(p;x)^{-2},} \tag{2.33}$$

so the analytic continuation of h follows once that of g is fixed by the pole-matching method.

3 Information-theoretic interpretation